Student Record System

C console app to store and display basic student records using arrays

Source Code (student_record.c)

```
/* Student Record System
   Stores and displays basic student records using arrays */

#include <stdio.h>
#include <string.h>

#define MAX_STUDENTS 50

struct Student {
    int rollNo;
    char name[50];
    float marks;
};

struct Student students[MAX_STUDENTS];
int studentCount = 0;

void addStudent() {
    if (studentCount >= MAX_STUDENTS) {
        printf("Student list is full.\n");
        return;
    }

    struct Student s;
    printf("Enter roll number: ");
    scanf("%d", &s.rollNo);

    printf("Enter name: ");
    scanf(" %[^\n]", s.name);

    printf("Enter marks: ");
    scanf("%f", &s.marks);

    students[studentCount] = s;
    studentCount++;
    printf("Student added successfully.\n");
}

void displayStudents() {
    if (studentCount == 0) {
        printf("No student records found.\n");
        return;
    }

    printf("\n%-10s %-20s %-10s\n", "Roll No", "Name", "Marks");
    printf("-----------------------------------------\n");
    for (int i = 0; i < studentCount; i++) {
        printf("%-10d %-20s %-10.2f\n", students[i].rollNo, students[i].name, students[i].marks);
    }
}

void searchStudent() {
    int roll;
```

```
    printf("Enter roll number to search: ");
    scanf("%d", &roll);

    for (int i = 0; i < studentCount; i++) {
        if (students[i].rollNo == roll) {
            printf("Found: %s, Marks: %.2f\n", students[i].name, students[i].marks);
            return;
        }
    }
    printf("Student with roll number %d not found.\n", roll);
}

int main() {
    int choice;

    do {
        printf("\n===== Student Record System =====\n");
        printf("1. Add Student\n");
        printf("2. Display All Students\n");
        printf("3. Search Student by Roll Number\n");
        printf("4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                addStudent();
                break;
            case 2:
                displayStudents();
                break;
            case 3:
                searchStudent();
                break;
            case 4:
                printf("Exiting program. Goodbye!\n");
                break;
            default:
                printf("Invalid choice, try again.\n");
        }
    } while (choice != 4);

    return 0;
}
```